

Relation Algebra

§9. $\frac{R}{S}$

$$R/\textcolor{teal}{S} \triangleq (R/S) \cap (S/R)^\circ$$

what R sends x to and nothing else is what S sends y to. In Rel x and y has the same image: $\forall p. (x \textcolor{teal}{R} p \iff y \textcolor{teal}{S} p)$

(9a)

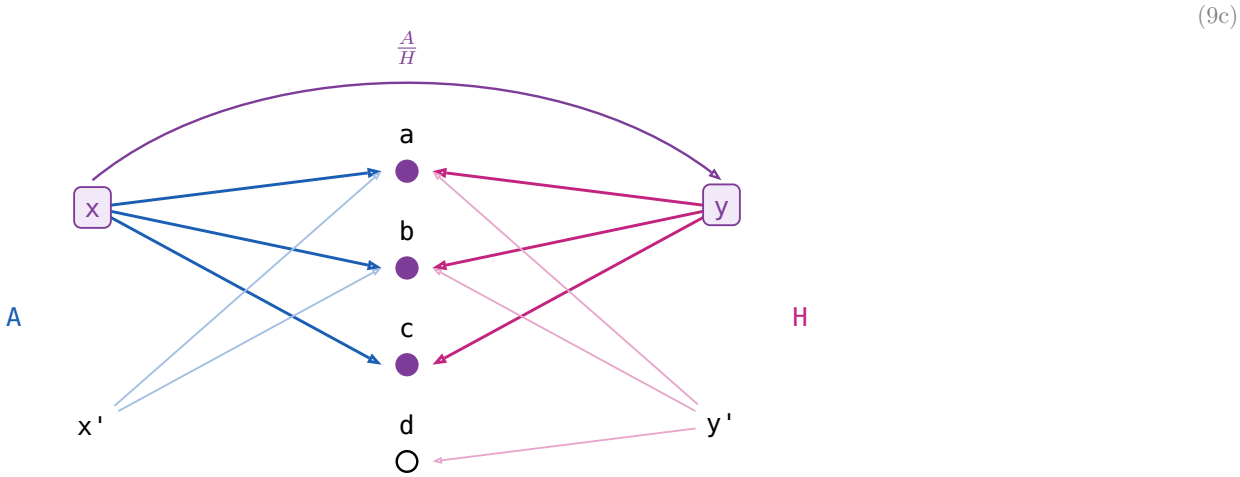
$$x \textcolor{teal}{(A/H)} y \text{ --- } x \text{ admires everyone } y \text{ hates}$$

$$x \textcolor{teal}{((H/A)^\circ)} y \text{ --- } x \text{ admires only people } y \text{ hates}$$

$$x \textcolor{teal}{((A/H) \cap (H/A)^\circ)} y \text{ --- } x \text{ admires only and all whom } y \text{ hates}$$

(9b)

x admires exactly whom y hates: $/$ is **all of**, $\frac{R}{S}$ is **only all of**.



law	the reading
$X \in R /_s S \iff X S \in R \wedge X \circ R \in S$ X sits inside the matching exactly when it carries S back inside R and, reversed, carries R back inside S .	X may pair x with y only when x admires exactly whom y hates. Both halves must typecheck, so the operation is partial .
$(R /_s S)^\circ = S /_s R$ reversing the matching swaps the two relations.	Matching is symmetric.
$(R /_s S) (S /_s W) \in R /_s W$ a match from R to S followed by one from S to W is a match from R to W .	And transitive.
$(R /_s S) S \in R$ following the matching by S lands inside R .	$(\exists y. x(\frac{R}{S})y \wedge ySp) \rightarrow xRp$ x only admires whom y hates $\frac{R}{S}S = \text{Dom}(\frac{R}{S})R$
$(R /_s R)R = R$ matching R against itself and then following R gives R back.	$(\exists y. x(\frac{R}{R})y \wedge yRp) \iff xRp$ x and y admire the same people $y=x$ always qualifies ($1 \in R \% R$ below)
$1 \in R /_s R$ every x matches itself.	$x(\frac{R}{R})y$ if x and y admires the same peoples.
$(R /_s R) (R /_s R) = R /_s R$ matching twice matches no more pairs than matching once.	So the relation admires the same people is an equivalence relation.
$\text{Symmetric}(X) \implies X \in R /_s R \iff X R \in R$ a symmetric X sits inside the matching exactly when following it by R adds nothing to R .	The largest symmetric arrow that leaves R alone.
$\text{simplePart}(R) \triangleq R /_s 1$ matching R against the identity keeps the x whose image is a single point.	The people who admire exactly one person and nobody else. It equals R only when R is simple, unlike $R/1 = R$.
$\text{dom}(R /_s S) = 1n(R/S) (S/R)$ the x that match something are those the two one-sided divisions take round the loop back to x .	Its domain is the domain of simplicity of R .